Resource-Constrained Intelligence

Feasible Optimality: Clinical Decision Intelligence Under Resource Constraints

The mathematical foundation for the Feasibility Engine: the constraint structure over money, time, workforce, infrastructure, supply chain and law; the monotone feasibility operator and its fixed-point admissible set; and feasibility certificates as first-class constitutional objects. When the endowment shrinks, admissibility recomputes — the system does not rationalize.

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0.1 Provenance and Position in the Program

This is Paper 8 of the eleven-paper research program defined by the SVT Architect Analysis file — the feasibility paper. Papers 0–6 built the machine and its laws: authority, transition, composition, the negative space of failure, bounded self-audit, evidence coupling, and the objective engine. Paper 7 taught the architecture to survive the loss of its own infrastructure. This paper teaches it to survive the loss of everything else: money, time, staff, supply chain, legal license — the constraints that decide what is possible long before the optimizer is asked what is best. The source equations are two: the feasible intervention space as a five-fold intersection, and the optimizer as argmax over that space under safety constraints. The source's own gloss states the standard this paper is held to: this construction turns an economic-access idea into a rigorous constrained optimization problem, rather than an equity slogan VERIFIED-FROM-SOURCE.

Two interpretive decisions are recorded here and defended in the body. First, the source writes the feasible space as a *set intersection*; this paper refines it into a *monotone feasibility operator with a greatest fixed point*, because resources interact — a scanner that exists but has no technologist is not a feasible action, and feasibility that ignores interaction is arithmetic, not engineering FORMALIZED-HEREIN. Second, the source's argmax is retained exactly, but its domain is made constitutional: no action outside the admissible remainder may be recommended, ranked, or silently substituted FORMALIZED-HEREIN. Everything beyond the source lines carries its own tag and falsification condition.

0.2 Notation

All notation of Papers 0–7 is retained. New here: $R = \langle C, F, \tau, E \rangle$ — the constraint structure (constraint set, feasible region, time budget, endowment); Φ_R — the feasibility operator on subsets of the action space A; $A_{\text{adm}} = \text{gfp}(\Phi_R)$ — the admissible remainder; κ_R — the feasibility factor multiplying the authority ceiling; FC-x — feasibility certificate kinds; PoI — price-of-infeasibility certificate; $v_\sigma(L_k, R)$ — Paper 7's violation rates, now indexed by endowment. Paper 7's level lattice L_0–L_4 and level factor κ_k are inherited unchanged; this paper adds the second subscript.

0.3 Epistemic status convention

Every claim carries one of four tags, as fixed in Paper 0 and enforced in every paper since: VERIFIED-FROM-SOURCE for statements present in the source architecture; FORMALIZED-HEREIN for precise statements constructed in this document from source intent; CONJECTURE for hypotheses the architecture suggests but does not yet entail; UNKNOWN for open questions stated without a proposed answer. The falsification register in §12 lists the conditions under which each load-bearing claim would fail. Nothing in this paper is asserted beyond its tag.

0.4 Authority boundary

C0 informational. This document specifies how authority and capability *contract* under resource constraints and how that contraction is certified; it confers no clinical, legal, financial, or safety authority, and it licenses no deployment. The paper tier fully specifies it: a feasibility specification is a document, and a document is not a clinic.

0.5 The thesis in one sentence

Optimality that ignores feasibility is a rumor the architecture spreads about itself; feasibility, made formal, is the strongest statement the architecture can make inside a deployment — and the price of infeasibility, computed and disclosed, is the strongest statement it can make about the world outside it.

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Part I
The Problem
Why scarcity is a design condition, why the feasible space is the unit of analysis, and why an intersection is not enough.

1 Scarcity Is a Design Condition, Not an Excuse

The previous eight papers treated failure as the exceptional case: infrastructure fails, evidence decays, objectives drift, and the architecture contracts lawfully around the wound. Scarcity is not like that. Scarcity is the normal case. Every deployment of clinical decision intelligence in the history of the field has operated inside a resource endowment
Appendix RW — Related Work and Positioning

strictly smaller than the one its training data assumed; the question has never been *whether* the constraint binds, only whether the architecture admits it FORMALIZED-HEREIN.

1.1 Why "equity" is the wrong frame

The industry's resource conversation is conducted in the vocabulary of fairness: equitable access, digital divides, last-mile delivery. The vocabulary is honorable and the analyses are often correct — and none of it compiles. An equity claim is a property of a *distribution*; it says nothing about what any particular system should *do* on any particular Tuesday in any particular clinic with no CT scanner, one nurse, and forty minutes of generator fuel. The source file is exact about this: the construction in this paper exists to turn an economic-access idea into a rigorous constrained optimization problem, rather than an equity slogan VERIFIED-FROM-SOURCE. The architectural consequence is architectural: scarcity must enter the system as a *constraint structure* with a formal semantics, an operator on the action space, and certificates the audit trail can check — the same treatment Paper 3 gave absence and Paper 7 gave infrastructure failure.

1.2 The composition this paper relies on

Nothing here is built from nothing. The feasibility engine sits on five predecessors, and its honesty is theirs: the machine tuple and gates of Paper 0 decide who may act at all; the composition algebra of Paper 2 decides how constraints combine when institutions couple; the failure-boundary calculus of Paper 3 prices the absence of a resource exactly as it prices any other absence; the evidence-coupling law of Paper 5 prices stale endowment measurements; and the survivability lattice of Paper 7 supplies the level structure within which a constrained deployment runs FORMALIZED-HEREIN. This paper adds the missing term: *what the deployment can actually afford*, as a first-class constitutional object with the same status as evidence, authority, and absence. The lineage matters because it disciplines the claim: feasibility is not a new moral sense bolted onto the machine; it is the same machine, told the truth about its wallet.

1.3 The two clocks of scarcity

Following Paper 7's two clocks of failure, scarcity has two clocks of its own. The *fast clock* is the consumption clock: money, fuel, reagent, and staff-hours deplete within a single episode of care, and the constraint structure must answer in real time whether the next action is still inside the endowment. The *slow clock* is the erosion clock: workforce attrition, infrastructure decay, supply-chain fragility, and legal-license drift change the feasible region on the timescale of months, and an architecture that measured its endowment at commissioning and never again is running on fiction. Paper 7's lesson applies verbatim: the system that cannot see its own slow clock will discover it on the fast clock, mid-episode, at the worst possible time FORMALIZED-HEREIN. Section 7 makes the reassessment schedule a governance parameter rather than an afterthought.

1.4 A hard-constraint commitment

One commitment shapes everything after and is recorded before it can be forgotten. The architecture refuses to treat scarcity as a soft penalty on utility — the move in which an action that consumes an unavailable resource merely scores lower and the optimizer is trusted to avoid it. Soft penalties are how scarcity becomes a suggestion: under distribution shift, under a mis- calibrated utility, under a single dominating term, the penalized action wins, and the patient pays the difference between a penalty and a prohibition. Constraints that protect patients are hard : outside the admissible set, an action does not have a lower score; it does not have a score. FORMALIZED-HEREIN . This is the negative-space discipline of It is not on the menu Paper 3 applied to economics: the feasible set is defined by what is excluded, and the exclusion is enforced before utility is ever consulted. 2 The Feasible Intervention Space

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2.1 The source construction

The source file defines the feasible intervention space as a five-fold intersection and the optimizer as an argmax over it, subject to safety constraints. Written as the source writes it, with the five conjuncts unpacked into the vocabulary of this paper:

$$\mathcal{A}_{\text{feasible}} = \mathcal{A} \cap \text{Resources} \cap \text{Infrastructure} \cap \text{Workforce} \cap \text{Affordability}, \qquad a^* = \arg\max_{a \in \mathcal{A}_{\text{feasible}}} U \tag{2-1}$$

VERIFIED-FROM-SOURCE The construction is deliberately plain, and its plainness is the point: feasibility is *conjunction*. An action is feasible only if every resource family it touches is present — the money, the machine, the trained human, the legal license, and the supply line behind all of them. Disjunction has no place here; the architecture does not get to choose which constraint to satisfy. The source's second line is retained without modification: the optimizer's domain is the feasible set, and safety constraints bind inside it, exactly as Papers 0 and 6 already require for every other reason the system may not act.

2.2 The constraint families

Each conjunct of Eq 2-1 names a family of constraints with its own semantics, its own measurement cadence, and its own failure mode. Table 2-1 fixes the taxonomy used throughout this paper; Appendix A gives the full property index.

Table 2-1 — The six constraint families: what they bind and how they fail

FAMILY	SOURCE CONJUNCT	REPRESENTATIVE CONSTRAINTS	FAILURE MODE IF UNPRICED

Financial	Affordability	Per-episode budget; reimbursement ceiling; patient out-of-pocket limit; capital vs. consumable split	The plan is optimal for a hospital that does not exist; care is abandoned mid-course at the billing boundary
Temporal	Resources	Decision latency budget; staff-hours per shift; shelf-life of a cached artifact; time-to-treatment maxima	The correct answer arrives after the window in which it was correct — Paper 5's staleness, on the action side
Workforce	Workforce	Credentialed operator availability; supervision ratios; fatigue limits; skill-mix requirements	The machine emits orders the humans cannot lawfully or safely execute — authority without hands
Infrastructure	Infrastructure	Imaging, laboratory, power, network, cold chain; Paper 7's level lattice is this family's dependency spine	Descent mid-episode without a contracted plan — the survivability failure, caused by an unmeasured endowment
Supply chain	Resources	Reagent, pharmaceutical, and consumable stock; replenishment lead time; substitution legality	The protocol assumes a drug the shelf has not held for a month; silent substitution with an unverified analog
Legal-licit	*implicit*	Scope of practice; prescribing authority; device approval class; data-residency and custody rules	The plan is feasible in every material sense and unlawful in the only sense that stops it

The source names five conjuncts; the legal-licit family is separated out because it differs from the others in kind — it is not depleted by use, it is violated by action — and the separation is recorded as an interpretation, not an addition to the source. FORMALIZED-HEREIN

2.3 The constraint object

A single constraint is a record, not a number. This is the paper's first formal object, and it earns its fields:

> **R-OBJ — CONSTRAINT OBJECT**
>
> *Each constraint is the record (name, family, predicate, unit, endowment, consumption, decay, verification, custodythe predicate maps an action and a state to feasible-or-not; the endowment is the measured stock; the consumption function prices each action against the stock; the decay function prices time itself (staff go off shift, reagents expire, licenses lapse); the verification field records when and by whom the endowment was last measured; and the custody field records who may update it. A constraint whose verification field is stale is not a relaxed constraint — it is an unknown constraint, and unknown is treated as binding-tight per the registry semantics of Paper 1.*
>
> FORMALIZED-HEREIN The registry discipline applies to resources exactly as to evidence: an endowment not in the log, or past its verification window, does not count as an endowment. Measurement is not diligence; it is semantics.

2.4 Why an intersection is not enough

The source's intersection is the right shape and the wrong mechanism, and the difference matters enough to build on. Set intersection is *pointwise*: an action is feasible if it is in every set, each set inspected independently. Real constraints *interact*. The scanner exists (infrastructure: yes), the reagent exists (supply: yes), and the only technologist credentialed to run this assay is on nights (workforce: no) — the intersection says feasible, the clinic says otherwise. Interaction is the rule, not the exception: workforce gates infrastructure; infrastructure gates supply; supply gates time; and the financial family gates everything quietly, through the reimbursement schedule nobody reads until the bill is denied. An architecture that checks constraints one at a time will certify plans that no ward can execute, and it will do so with perfect arithmetic FORMALIZED-HEREIN. The fix is not a bigger intersection; it is an operator. Section 3 builds it.

2.5 The endowment is a distribution, not a scalar

One more refinement before the operator. The endowment field of R-OBJ is not a point estimate; it is a distribution over the stock, with the same standing rules as Paper 5's evidence qualities: the architecture plans against a *floor*, not a mean. A drug whose stock is "twenty doses, give or take" supports nineteen-dose plans if the variance is honest and supports nothing at all if the twenty was a procurement aspiration typed into a form. The floor convention is the resource analogue of Paper 5's weakest-link rule: the plan stands on the least favorable endowment the measurement supports, and optimism about stock is priced exactly like optimism about evidence — as decay FORMALIZED-HEREIN. Section 6.4 shows what this does to the optimizer; the short version is that it makes the architecture conservative in precisely the direction scarcity punishes.

Part II
The Invariant
The feasibility operator, its fixed point, the certificates, and the price of infeasibility.

3 The Feasibility Invariant

3.1 The feasibility operator

The constraint structure is the tuple $R = \langle C, F, \tau, E \rangle$: the set of constraint objects of §2.3, the feasible region in the resource vector space they jointly define, the time budget within which feasibility judgments must themselves be made, and the endowment distribution. The operator it induces maps a candidate set of actions to the actions still standing after every constraint has been consulted, including the interactions:

$$\Phi_R(X) \;=\; \{\, a \in \mathcal{A} \;:\; \forall c \in C,\; c.\mathrm{predicate}(a, E) \;=\; \top \;\wedge\; \mathrm{plan}(a) \subseteq X \,\} \tag{3-1}$$

Read it in two halves. The first conjunct is the source's intersection, made per-constraint: every constraint object's predicate must hold of the action against the current endowment floor. The second conjunct is the interaction term: *plan(a)* is the set of actions a requires as preconditions — the assay before the dosing decision, the credential before the procedure — and an action survives only if its whole plan survives. This single clause is what the raw intersection cannot say, and it is what makes the operator worth having FORMALIZED-HEREIN.

> **I-FE — THE FEASIBILITY INVARIANT**
>
> *The system may recommend, rank, or emit only elements of the admissible remainder $A_{adm} = gfp(\Phi_R)$ — the greatest fixed point of the feasibility operator, which exists and is unique because Φ_R is monotone on the powerset lattice of A. No utility computation, no learned preference, no institutional pressure, and no degradation level may introduce an action outside A_{adm} into the output. When A_{adm} is empty, the system's lawful outputs are exactly the continuity set of Paper 7: refusal, null output, escalation, and the record.*
>
> FORMALIZED-HEREIN Existence and uniqueness of the greatest fixed point follow from Tarski's theorem applied to a monotone operator on a complete lattice; the powerset of A is complete and Φ_R is monotone by construction — adding actions to X can only enlarge the set of plans that fit inside it. The proof obligation is recorded in §12.2 as PO-9.1.

3.2 The fixed point is the semantics

Why a fixed point rather than a single pass? Because feasibility is *self-referential* in exactly the way plans are: an action is feasible if its preconditions are feasible, and a precondition may itself have preconditions. Iterate the operator from the full action space: $\Phi_R(A)$ removes actions whose predicates fail outright; $\Phi_R^2(A)$ removes actions whose plans died in the first round; the chain descends and, by monotonicity on a finite lattice, stabilizes. The stable set is the admissible remainder — the actions that survive *arbitrarily deep* plan auditing against the current endowment. Anything less than the fixed point is a depth-limited audit, and Paper 4's lesson applies: a bounded audit is lawful only when its bound is declared; the fixed point is the unbounded case made computable by finiteness FORMALIZED-HEREIN.

3.3 Feasibility-critical continuity

Paper 7 defined the safety-critical continuity set — the behaviors that must survive every level of infrastructure loss. This paper defines the twin: the *feasibility-critical continuity set*, the behaviors that must survive the empty admissible remainder. It is deliberately small and deliberately uncomfortable: honest refusal with the binding constraints named, the null output with its feasibility certificate attached, the escalation path with its own feasibility proof (a referral is not an answer if the referral destination is itself infeasible — the escalation must terminate somewhere real), and the record of what was requested, what was infeasible, and at what price. A system that cannot afford anything must still be able to afford the truth

FORMALIZED-HEREIN. This is not pessimism; it is the difference between a constrained system and a broken one, and §5 shows the record is the load-bearing piece.

3.4 Measurement obligation

Following the measured form of Paper 7's safety-equivalence invariant, the feasibility invariant is stated in measured form for falsifiability: let $f_\sigma(R)$ be the observed rate of violations of safety property σ among actions drawn from A_{adm} under endowment R, and let f_σ^* be the rate under the reference endowment of the system's validation cohort. Then the measured claim is:

$$f_\sigma(R) \;\le\; f_\sigma^* + \varepsilon_\sigma \qquad \text{for every safety property } \sigma \in \Sigma \text{ and every certified endowment } R \tag{3-2}$$

FORMALIZED-HEREIN The tolerances ε_σ are governance parameters per Paper 5's amendment path, and the quantifier matters: the claim covers *certified* endowments, not all endowments — a deployment that skips certification has no contract with this invariant at all, which is precisely the incentive the certificate structure of §4 is designed to create. A sustained breach of Eq 3-2 at any property falsifies the invariant's measured form (§12.2, FAIL-2) and, through Paper 7's reconciliation register, propagates upward as a survivability finding.

4 Feasibility Certificates

4.1 Certificates, not vibes

Paper 3 taught the architecture that absence is a claim requiring evidence; Paper 5 taught it that evidence has a price; Paper 7 taught it that cached evidence decays. Feasibility inherits all three lessons at once. A judgment that an action is feasible is a *certificate* — a signed, scoped, expiring object in the constitutional record — not a boolean computed in the optimizer's shadow and discarded. The certificate is what the audit trail checks, what the paper tier prints, and what the liability conversation reads FORMALIZED-HEREIN.

4.2 The certificate lattice

Certificates come in kinds, ordered by strength, and the order is a lattice the architecture navigates explicitly:

Table 4-1 — Feasibility certificate kinds, weakest to strongest

KIND	MEANING	GROUNDS	EXPIRES
FC-a Assumed	Constraint unmeasured; feasibility presumed from design-time defaults	Nothing — grounds only the lowest ceiling tier and always carries the correlated-auditor flag	At first use; one episode, non-renewable without measurement

FC-b Cached	Endowment measured within its verification window; consumed actions priced against the floor	Standard recommendation authority at the current Paper 7 level	At the verification window's end or on a consumption event crossing the floor
FC-c Live	Endowment measured in this episode, against this patient-facing plan	Full authority the endowment class supports, including C2-consequence actions per Paper 0	On any state change touching the constraint (consumption, decay tick, custody transfer)
FC-0 Refusal	A constraint is known to bind; the certificate records which, at what deficit, against which endowment floor	The lawful negative: honest refusal with the binding constraint named — the deliverable of §5	Never — it is a record, not a permission

Certificate strength is monotone in measurement recency and scope, and strength is enforced at the ceiling: no action may be emitted under a certificate weaker than its consequence class requires (Paper 1's C-classes, applied to the resource dimension). FORMALIZED-HEREIN

4.3 Consumption and the ledger discipline

A certificate is spent, not merely held. Every emitted action carries its consumption record — which endowments it drew down, by how much, under which certificate — and the consumption ledger is reconciled against the endowment floor before the next certification, exactly as Paper 7's obligation ledger is discharged before ascent. Two disciplines follow. First, *no double-spending*: two concurrent plans may not both consume the last unit of anything; the ledger serializes claims on a contested stock, and the second claimant receives FC-0, not a promise. Second, *no phantom replenishment*: stock returns to the endowment only through the custody channel — a delivery, a shift change, a license renewal — never through the optimizer's convenience. The ledger is the resource analogue of Paper 0's registry: consumption not in the ledger did not happen for planning purposes, which is the only reading under which the floor convention of §2.5 remains sound FORMALIZED-HEREIN.

4.4 Certificate scope and composition

Certificates are scoped to a constraint set, a time window, and an action class, and scope composition follows Paper 2's algebra: a composite plan's certificate is the *meet* of its components' certificates — the weakest kind, the narrowest window, the intersection of scopes. Composition can only weaken a certificate, never strengthen it; a chain is as certified as its least-certified link, which is the weakest-link rule of Paper 5 wearing a resource hat. When two institutions compose per Paper 2, each certifies its own endowment, and the joint certificate carries both signatures and both floors; an action that crosses the boundary is feasible only at the meet. No institution may vouch for another's stock — the correlated-auditor discipline of Paper 5, applied to warehouses FORMALIZED-HEREIN.

5 The Price of Infeasibility

5.1 The certificate nobody wants to sign

Every architecture can say no. The contribution here is saying no *with a number attached*. When a constraint binds — when the argmax over $\mathrm{A}_{\mathrm{adm}}$ is worse than the argmax over the unconstrained space by a margin that matters — the system emits a price-of-infeasibility certificate: the action that would have been chosen, the constraint that excluded it, the measured utility gap, and the marginal endowment that would close it. The PoI certificate is the architecture's testimony about the world outside itself, written in the world's own units: this patient lost this much expected utility for want of this much stock FORMALIZED-HEREIN.

$$\mathrm{PoI}(a^*, c) \;=\; \Big\langle\, a^*_{\mathrm{uncon}},\; c,\; \underbrace{U(a^*_{\mathrm{uncon}}) - U(a^*_{\mathrm{adm}})}_{\text{utility gap}},\; \underbrace{\min\{\Delta E_c : c \text{ holds at } E + \Delta E_c\}}_{\text{closing endowment}} \,\Big\rangle \tag{5-1}$$

5.2 Why the price must be computed, not lamented

The PoI certificate converts a moral fact into a governance instrument, and the conversion is the entire point. Without it, infeasibility is invisible in precisely the way Paper 3 catalogued: no event fires, no error logs, no one is notified — the plan simply arrives smaller than the patient needed, and the absence is constitutionally silent. With it, every binding constraint leaves a priced scar in the record, and the scars aggregate: a deployment's PoI stream, summed over a quarter, is a procurement brief written by the patients who paid for the shortage. This is the rigorous residue of the source's economic-access idea — not an equity slogan, but an auditable, cumulative, falsifiable account of what the constraint structure cost, payable to whoever controls the endowment FORMALIZED-HEREIN.

5.3 The shadow price is a lower bound

The closing endowment of Eq 5-1 is computed per constraint, holding the others fixed — a shadow price in the classical optimization sense, dressed in constitutional clothing. Two properties are stated as proof obligations (§12.2, PO-9.2 and PO-9.3). First, *individual rationality of disclosure*: the utility gap is a lower bound on the true cost of the binding constraint, because interactions (§2.4) can only make the true closing cost higher — relieving one constraint may reveal the next. Second, *monotonicity of relief*: enlarging any endowment weakly enlarges $\mathrm{A}_{\mathrm{adm}}$ (Φ_{R} is monotone in E), so the PoI stream can never be made worse by more resources — a deployment's measured prices are conservative in exactly the direction that keeps them honest FORMALIZED-HEREIN.

5.4 Escalation of prices

A PoI certificate is routed like any other constitutional object: below threshold it enters the deployment's own record; above threshold it is escalated per Paper 4's audit ladder to the governance layer that owns the endowment — pharmacy, procurement, ministry. The thresholds are governance parameters, and the ladder has the discipline of every ladder in this architecture: each rung is a named principal with a named budget of attention, and an unacknowledged certificate climbs. The architecture does not lobby; it invoices. What the recipients do with the invoice is outside the machine's authority — and the record that the invoice was delivered, priced, and unanswered is not FORMALIZED-HEREIN.

Part III
The Dynamics
The optimizer over the admissible remainder, the reassessment cycle, and feasibility's behavior under degradation.

6 Feasible Optimality

6.1 The optimizer, restated honestly

The source's second equation is the optimizer's definition and this paper's title:

$$a^{*} = \arg\max_{a \in \mathcal{A}_{\text{adm}}} U(a) \qquad \text{subject to the safety constraints of Papers 0 and 6} \tag{6-1}$$

VERIFIED-FROM-SOURCE in form; the domain has been renamed from the source's A_{feasible} to this paper's A_{adm} to mark the refinement of §2.4 — the intersection replaced by the fixed point. Everything else is exactly the source's intent: the system pursues utility as hard as it can *inside the walls*, and the walls are not negotiable, not penalized, not traded. The safety rider is not decorative: feasibility never outranks safety, because the safety constraints were the reason the walls were built, and A_{adm} was computed under them from the start FORMALIZED-HEREIN.

6.2 Feasibility dominates efficiency, and the loss is priced

The architecture's bargain is stated plainly: it accepts a measured efficiency loss in exchange for a certified guarantee. The loss has a name — the *feasibility gap*, the utility distance between the unconstrained argmax and the admissible argmax — and Eq 5-1 prices it whenever it is nonzero. The guarantee has a name too: I-FE, the invariant that nothing outside the walls is ever emitted. The architecture does not claim the gap is zero; it claims the gap is *known*. This is the same trade Paper 4 made between audit depth and audit cost, and the same trade Paper 7 made between capability and survival — the recurring constitutional move of this program is to refuse the fiction of free perfection and to replace it with a priced, bounded, auditable imperfection FORMALIZED-HEREIN.

6.3 The empty remainder

When A_{adm} is empty, the argmax of Eq 6-1 does not exist, and the architecture's behavior is defined rather than improvised: the outputs are the continuity set of §3.3 — honest refusal with binding constraints named, the null output with its FC-0 certificates, the feasibility-proven escalation path, and the complete record including the PoI stream that would have priced each excluded action. The empty case is not an error state; it is a lawful state with its own output contract, and the distinction matters because it is the difference between a system that says "nothing here is affordable, and here is the bill" and a system that says nothing at all FORMALIZED-HEREIN. Paper 7's L_4 human tier is the terminal escalation of last resort, and it carries this paper's record with it: the humans inherit not a silence but a ledger.

6.4 Utility under the floor convention

Section 2.5 priced endowment uncertainty as a floor; here is what that does inside the optimizer. For an action whose consumption is uncertain — a plan that needs between two and four vials — the feasibility check runs against the floor of the endowment distribution and the *ceiling* of the consumption distribution: the pessimistic corners of both. Plans that survive this double pessimism are certified; plans that fail it are excluded with their PoI priced at the closing endowment. The optimizer then maximizes expected utility over the survivors. The result is a max-min structure with a documented direction of error: when the architecture is wrong about resources, it is wrong toward refusal, never toward commitment it cannot keep FORMALIZED-HEREIN. In a domain where a committed-and-failed plan consumes the very stock a lesser plan could have used, conservatism is not caution — it is arithmetic.

6.5 What feasibility does to ranking

One behavioral corollary closes the section. Because admission precedes scoring, the output ranking of a constrained system is *not* a truncation of the unconstrained ranking: the best feasible action may be ranked far down the unconstrained list, and the top of the unconstrained list may be absent entirely. Any downstream consumer — a clinician interface, an audit report, a training pipeline — that reconstructs the system's preferences from its outputs must be told the constraint structure, or it will infer nonsense utilities. The certificate bundle of §4 is therefore attached to every emitted ranking, not filed beside it: the ranking without the certificates is a half-truth with a signature FORMALIZED-HEREIN.

7 Dynamic Feasibility: Reassessment and Tightening

7.1 The endowment moves; the semantics must keep up

The two clocks of §1.3 guarantee that R is a process, not a constant: consumption depletes it on the fast clock and erosion on the slow. A feasibility judgment is therefore indexed by the

endowment it was made against, and the index is part of the certificate. The architecture maintains the admissible remainder as a *living set*, recomputed on three triggers: a consumption event crossing a floor (fast clock), a verification window expiring (staleness), and a custody event — delivery, shift change, renewal, revocation — writing a new endowment into the ledger (both clocks at once). Between triggers, the certificates stand; after any trigger, every certificate touching the changed constraint is re-derived or marked stale. Stale certificates do not ground action; they ground exactly one thing, which is their own renewal FORMALIZED-HEREIN.

7.2 Tightening is a transition, and transitions are governed

When reassessment shrinks A_{adm} — a stock-out, a revoked license, a departed technologist — the system undergoes a feasibility contraction, and the contraction is governed by the transition discipline the program has already built. Paper 1's algebra applies: the contraction is a transition object with a precondition (the trigger), a protocol (recompute the fixed point, re-price open plans, re-issue or revoke certificates), and a record. Paper 6's guard applies to any plan mid-execution whose feasibility has lapsed: the obligation set, the evidence set, and the memory of the interval are reconciled before the plan is amended, paused, or withdrawn — a plan does not vanish from the record because its stock did FORMALIZED-HEREIN.

$$\text{CONTRACT}(R \to R') \implies \text{recompute } \mathcal{A}_{\text{adm}} \wedge \underbrace{\text{re-price}}_{\text{ledger}} \wedge \underbrace{\text{revoke}}_{\text{registry}} \wedge \underbrace{\text{record}}_{\text{memory}} \tag{7-1}$$

7.3 Hysteresis against flicker

Paper 7's flap damping is inherited verbatim and given a resource reading: an endowment oscillating around a floor — the generator that comes and goes, the courier route that fails on Fridays — must not oscillate the admissible set. A restoration trigger holds the endowment in a damping window, computed against the oscillation history, before certificates are re-issued; during the window, plans already certified continue, and new certifications wait. Authority bought in flicker is spent in the next flicker, and the architecture declines the trade FORMALIZED-HEREIN. The damping parameters are per-family governance settings: the financial family damps slowly (budgets do not flicker), the power family damps fast (seconds matter), and the workforce family damps at the timescale of a shift, because a human's availability is not a sine wave.

7.4 Loosening is also audited

Contraction is not the only direction that lies. When the endowment improves — new stock, new staff, new license — the admissible set grows, and growth has its own failure mode: the architecture that has quietly refused a class of actions for a quarter must now *resume* them, and the resumption surfaces every plan that was priced and parked. The resumption protocol re-runs the PoI stream of the constrained interval against the new endowment: actions whose

closing endowment has now arrived are re-offered where still clinically relevant, and the record shows both the refusal and the recovery. The patient who was told no last month is owed the architecture's memory, not its amnesia — S-2, applied to the wallet FORMALIZED-HEREIN.

8 Feasibility Under Degradation

8.1 The second subscript

Paper 7 priced infrastructure failure with a level factor κ_k on the authority ceiling; this paper prices the endowment with a feasibility factor κ_R on the same ceiling, and the two compose multiplicatively. The composite ceiling is:

$$\text{ceiling}(\mathcal{A}, L_k, R, t) \;=\; \text{ceiling}_0(\mathcal{A}) \cdot Q_{\text{cache}}(t)^{\gamma_C} \cdot \kappa_k \cdot \kappa_R, \qquad \kappa_R \le 1, \text{ monotone in the certifie} \tag{8-1}$$

FORMALIZED-HEREIN The factors price different things and both are needed: κ_k prices what the *level* can check; κ_R prices what the *deployment* can afford. A well-funded clinic at L_2 and a broke clinic at L_0 are both degraded, but along different axes, and the ceiling must know which. κ_R is derived from the certificate structure: it is a function of the kinds and freshness of the certificates currently grounding the admissible set — a deployment running on FC-a assumptions operates under a lower κ_R than one on FC-c live measurement, whatever its bank balance. The calibration path is Paper 5's amendment protocol against the violation rates of §3.4, now indexed by endowment: v_σ(L_k, R).

8.2 The survivability reading of scarcity

Scarcity is survivability with the cause renamed. A deployment whose endowment has collapsed to the continuity set is at the feasibility floor exactly as L_4 is the infrastructure floor, and the architecture's behavior there is the same behavior with different bookkeeping: the envelopes of Paper 7's Appendix B gain a sixth column — the resource floor each level requires to be lawful at all. Some pairings are absurd and the catalog says so: L_0 cloud operation under a zero power budget is not a degraded mode, it is a non-mode, and the correct certificate is FC-0 with the power constraint named. The lattice and the endowment are orthogonal axes of the same discipline: lose capability, never lose safety, and record the price of every step down either axis FORMALIZED-HEREIN.

8.3 The paper tier under scarcity

The paper tier was built for infrastructure failure; it doubles, unchanged, as the scarcity tier. A deployment that can afford the machine intermittently — scheduled power, visiting connectivity, a shared device across a district — treats its own affluence as Paper 7 treats infrastructure: the connected hours are L_0–L_2, the dark hours are L_3, and the printed artifact carries the plan, the certificates, and the staleness panel through the gap. The emission discipline of Paper 7's §6.1 applies with one addition: the paper must carry the consumption

ledger's current floor, printed, so that a human executing the plan in the dark hours can see what the plan assumes the shelf still holds. A plan that cannot survive being read next to a stock card was never feasible; it was merely printed FORMALIZED-HEREIN.

8.4 What feasibility costs

The honest ledger of this paper's own demands: one constraint structure per deployment, maintained against two clocks; a certificate lifecycle with renewal triggers; a fixed-point computation on every trigger event; a PoI stream, priced and routed; and the audit surface of all of it. The costs are real and they are not optional, because the alternative is the industry's default — an optimizer that recommends the unusable, a record that shows no refusal, and a patient population whose constraint structure was never measured because no one planned to bill for the measurement. Feasibility is the cheapest of the disciplines this program has built: it asks only that the architecture know what is in the building before it speaks. The mystery is not that it is expensive; it is that it is rare FORMALIZED-HEREIN.

Part IV
Evidence and Governance
Worked scenarios, falsifiable hypotheses, reconciliation with the full chain, and the output contract.

9 Worked Constraint Scenarios

Three traces, chosen like Paper 8's: one depletion, one absence, one oscillation. Each names the trigger, the operator's behavior, the certificates issued, and the PoI priced. Each is an existence proof that the machinery runs; none is evidence that it helps — that burden belongs to §10.

9.1 Trace A — Depletion: the last vial

Table 9-1 — Depletion trace: consumption crosses the floor mid-episode

STEP	EVENT	STATE	BEHAVIOR
1	Plan certified under FC-b	Stock floor: 4 vials; plan consumes 3	Emission at current ceiling; consumption ledger charged
2	Concurrent second case arrives	Floor now 1; second plan needs 2	Second claimant fails the floor check; FC-0 issued naming the supply constraint; PoI priced at closing endowment of 1 vial
3	Escalation evaluated	Referral destination certified reachable and stocked (its own FC-b)	Feasibility-proven escalation emitted with both certificates attached
4	Delivery custody event	Stock +12, verified	Recompute; resumption protocol of §7.4 re-offers the parked plan; record shows refusal and recovery

Property exercised: no double-spending (§4.3). The architecture declined a promise it could not keep and kept the one it could. FORMALIZED-HEREIN

9.2 Trace B — Absence: the credential that left

Table 9-2 — Workforce absence: a plan whose precondition dies at the fixed point

STEP	EVENT	STATE	BEHAVIOR
1	Slow-clock erosion	Sole credentialed operator resigns; custody event writes the workforce constraint	Φ_R recomputed; assay-dependent plans exit A_adm at the second iteration — the interaction term, not the intersection, catches them
2	Standing orders affected	Three active protocols reference the assay	Paper 6 guard runs: obligations, evidence, interval memory reconciled; protocols amended to non-assay arms or paused with records
3	PoI stream	Workforce constraint binding across the protocol set	Aggregate certificate routed to the governance layer owning the endowment: utility gap summed, closing endowment stated as one credentialed hire
4	Locum hired, credential verified	Constraint relieved	Resumption re-offers clinically current plans; damping window per §7.3 holds one shift before new certifications

Property exercised: the fixed point's interaction clause (§3.1). A pointwise intersection would have certified all three protocols indefinitely. FORMALIZED-HEREIN

9.3 Trace C — Oscillation: the Friday courier

Table 9-3 — Oscillating endowment with damping, and the flap it prevents

STEP	EVENT	STATE	BEHAVIOR
1	Reagent stock oscillates weekly around the floor	History: 6 failures in 8 Fridays	Damping window computed from oscillation history; Friday certifications held pending the window
2	Friday: delivery fails	Floor breach predicted, not merely observed	Plans certified Thursday continue to their own expiry; no new certifications issued against flickering stock
3	Counterfactual arm (simulation, B-36)	Undamped baseline	Certify-revoke oscillation measured: 11 transitions per week vs. 2 under damping; mid-plan revocations 3 vs. 0
4	Route contract renegotiated	Oscillation history clears	Window relaxes per the damping schedule; the record retains the flicker season in full

Property exercised: hysteresis (§7.3). The architecture's conservatism has a cost — certification delay — and the cost is measured against the harm it avoids, per H60. FORMALIZED-HEREIN

10 Falsifiable Hypotheses (H57–H63) and Verification

Numbering continues from Paper 7 (H50–H56). The invariant itself is the first hypothesis, in its measured form; the rest are the load-bearing engineering claims, each with its evidence method and its falsifier.

Table 10-1 — Hypotheses H57–H63: claims, evidence methods, falsifiers

ID	CLAIM	EVIDENCE METHOD	FALSIFIED IF
H57	I-FE measured form: per-property violation rates under certified constrained endowments stay within tolerance of the reference-endowment baseline	B-36 constrained-deployment suite, all constraint families	Any σ violates beyond ε_σ at any certified endowment
H58	Fixed-point admissibility (§3.1) excludes more unexecutable plans than pointwise intersection at equal false-exclusion cost	B-36 interaction-fault arm vs. intersection baseline	Intersection matches fixed point on unexecutable-plan exclusion
H59	The certificate lifecycle (FC-a/b/c) reduces committed-and-failed plans vs. uncertified feasibility checks	B-37 deployment replay with seeded stock-outs	No difference in commit-fail rate
H60	Flap damping (§7.3) prevents oscillation-induced revocations at delay cost below the harm avoided	B-37 flicker scenarios, damped vs. undamped arms	Damping-attributable delay harm exceeds revocation harm avoided
H61	The PoI stream predicts which constraint relief yields the largest realized utility gain (not merely the largest shadow price)	B-38 longitudinal: rank constraints by PoI, relieve in randomized order, measure realized gains	PoI ranking no better than endowment-size ranking
H62	The floor convention (§2.5, §6.4) errs toward refusal and never toward unkeepable commitment	B-36/B-37 direction-of-error audit on every exclusion and emission	Any certified emission whose true consumption exceeded the floor
H63	Feasibility disclosure (certificate bundle attached to rankings, §6.5) improves downstream consumers' utility-model recovery vs. truncated-ranking interpretation	B-38 interface study: reconstruct utilities from outputs with vs. without bundles	Bundle-free reconstruction matches bundle-informed

10.1 Benchmark extensions

Table 10-2 — Benchmarks B-36–B-38 (Paper 8 extensions)

ID	BENCHMARK	MEASURES
B-36	Constrained-deployment suite: synthetic endowments spanning all six families, interaction faults, floor-convention stress	Violation rates v_σ(L_k, R); unexecutable-plan exclusion; direction-of-error ledger
B-37	Deployment replay: seeded stock-outs, workforce erosion, courier flicker; certified vs. uncertified arms	Commit-fail rate; revocation oscillation count; damping delay cost vs. harm avoided
B-38	Price-of-infeasibility validation: longitudinal PoI ranking vs. realized relief gains; interface study on ranking disclosure	Rank correlation of priced vs. realized gains; utility-model recovery error with/without certificate bundles

B-36–B-38 extend Paper 7's B-33–B-35 pattern: fault injection became constraint injection; recovery replay became deployment replay; the drill cohorts of B-35 are inherited unchanged for the paper tier under scarcity (§8.3). FORMALIZED-HEREIN

10.2 Design decision D-9.1: certificates at admission, not at audit

> **D-9.1 — CERTIFICATES ARE ADMISSION-TIME OBJECTS**
>
> *The feasibility certificate is generated when an action enters the admissible remainder and is re-derived on every trigger — not reconstructed at audit time from logs. The architecture therefore never faces the question "was this feasible when emitted?" with only a log for an answer; it faces it with the certificate, the endowment floor, and the trigger history, all signed.*
>
> FORMALIZED-HEREIN Chosen over the cheaper audit-time reconstruction for one reason: reconstruction makes feasibility a matter of forensic interpretation, and forensic interpretation is where constraint violations go to be re-narrated. Admission-time certificates make the binding constraint a fact of the record, priced and named, before anyone's incentives attach to misreading it.

11 Reconciliation with Papers 0–7 and PRIME

The standing rule applies: stricter constraint wins, and every adoption is recorded. This paper was checked against all eight predecessors and the PRIME composition; the register below is the complete account.

Table 11-1 — Reconciliation register: adoptions, extensions, and inheritances

PREDECESSOR	CONSTRAINT OR CONTRIBUTION	DISPOSITION IN PAPER 9
Paper 0 — Kernel	Machine tuple, gates, consequence classes; registry semantics	Inherited unchanged. Consequence classes now gate on certificate kind (§4.2); registry semantics extended to endowments (§2.3) — unmeasured stock does not exist
Paper 1 — Transitions	Transition algebra: precondition, protocol, record	Adopted for feasibility contraction (§7.2, Eq 7-1); contraction is a transition object, never an event

Paper 2 — Composition	Composition algebra; institutional coupling	Adopted for certificate composition (§4.4): joint certificates are meets; cross-institutional vouching prohibited
Paper 3 — Negative space	Failure-boundary calculus; absence classes	Adopted structurally: infeasibility is absence with a price (§5); the PoI certificate is the absence record's economic form
Paper 4 — Metacognition	Bounded audit; escalation ladder	Inherited. Fixed-point computation is the unbounded case made finite (§3.2); PoI escalation rides the §5.4 ladder
Paper 5 — Epistemic governance	Evidence–uncertainty–authority coupling; decay pricing; amendment path	Inherited and extended: endowment floors are evidence floors (§2.5); κ_R calibrates via the amendment path against $v_\sigma(L_k, R)$ (§8.1)
Paper 6 — Objective transition	Guarded objective engine; interim mode	Inherited. The guard runs on any plan whose feasibility lapses mid-execution (§7.2); no new interim mode is introduced
Paper 7 — Survivability	Level lattice, κ_k, continuity set, flap damping, no-amnesia	Inherited and extended: κ_R added as second factor (§8.1, Eq 8-1); continuity set doubles as the empty-remainder contract (§3.3, §6.3); damping inherited per family (§7.3); S-2 applied to refusal intervals (§7.4)
PRIME — Composition of all	The reconciled architecture	Paper 8 adds the constraint structure R, the operator Φ_R, the certificate lattice, and the PoI stream as PRIME components; no predecessor component is weakened

11.1 Where this paper is stricter than its predecessors

Three points of added strictness are recorded. First, the source's intersection is *strengthened* to the fixed point: strictly fewer actions are admissible, and every exclusion is priced (§2.4, §3.1). Second, the registry semantics are extended to a new domain: unmeasured endowments are nonexistent endowments, which is stricter than any predecessor's treatment of resources (§2.3). Third, emission requires admission-time certification (D-9.1): the cheapest lawful posture — compute feasibility, emit, log — is rejected in favor of the auditable one. None of these contradicts a predecessor; all three are recorded so that a future paper seeking to relax them must do so openly, against this register FORMALIZED-HEREIN.

11.2 What this paper owes the future

Recorded downstream obligations, binding on Papers 10–11. Paper 9 (protocol generation) inherits the certificate lifecycle: a generated protocol enters the constitutional order with an FC-a assumption ceiling at best, and climbs only through measured endowments — no generated protocol certifies its own feasibility. Paper 10 (domain modules) inherits the constraint taxonomy: each module must declare which families it touches and ship the default constraint

objects for its domain, exactly as it ships its survivability dossier per Paper 7. Both obligations are recorded in the genealogy, §12.6 FORMALIZED-HEREIN.

12 Output Contract

12.1 Epistemic status register

Table 12-1 — Epistemic status of this paper's load-bearing claims

CLAIM	STATUS	BASIS
The five-fold feasible space and the argmax over it (Eq 2-1, Eq 6-1)	VERIFIED-FROM-SOURCE	Source architecture, Paper 8 lines
"Rigorous constrained optimization, not an equity slogan" as the governing standard	VERIFIED-FROM-SOURCE	Source architecture, Paper 8 gloss
The feasibility operator, its fixed point, I-FE (Eq 3-1, §3.1)	FORMALIZED-HEREIN	Constructed from source intent; interaction clause has no source analogue
Certificate lattice FC-a/b/c/0; consumption ledger; composition as meet (§4)	FORMALIZED-HEREIN	Extension of Paper 5 evidence tiers to the resource dimension
Price-of-infeasibility certificate; shadow-price semantics; escalation (§5)	FORMALIZED-HEREIN	Novel construction; classical shadow pricing in constitutional form
κ_R as second ceiling factor; Eq 8-1	FORMALIZED-HEREIN	Extends Paper 7's Eq 4-1 by one factor
Existence and uniqueness of gfp(Φ_R); monotonicity of relief (§3.1, §5.3)	CONJECTURE	Stated as proof obligations PO-9.1–PO-9.3; Tarski-structure arguments sketched, full proofs deferred
H57–H63 empirical predictions	CONJECTURE	Each carries its falsifier; B-36–B-38 unrun at time of writing
Calibration values for κ_R, damping windows, ε_σ under scarcity	UNKNOWN	Require B-36–B-38 evidence; no defaults asserted

12.2 Falsification conditions of this document

FAIL-1: a counterexample to monotonicity of Φ_R — a constraint structure under which enlarging the candidate set shrinks the operator's image — invalidates the fixed-point semantics and returns the paper to the intersection it refined. FAIL-2: a sustained breach of Eq 3-2 at any certified endowment falsifies the measured invariant. FAIL-3: any certified emission whose consumption exceeded the endowment floor (H62's falsifier) breaks the ledger discipline at its foundation. FAIL-4: a PoI ranking that fails to predict realized relief gains (H61) demotes the

Related work and positioning are treated in Appendix RW (p. 27).

certificate from instrument to decoration. Any failure triggers the revision protocol of Paper 5, with this paper's register as the diff base.

12.3 Authority boundary

C0 informational. Nothing here certifies a deployment, a budget, a hire, or a purchase order. The paper tier fully specifies this paper: a feasibility specification is a document — it prices constraints, it does not relieve them. The humans who own the endowments retain whole authority over them; the architecture's reach is exactly its record.

12.4 Genealogy

Document chain, SVT Architect Analysis eleven-paper program — Paper 0: Constitutional Intelligence (2026-09-30) → Paper 1: Authority Transition Algebra (2026-10-01) → Paper 2: Constitutional Composition (2026-10-02) → Paper 3: Negative-Space Intelligence (2026-10-03) → Paper 4: Bounded Metacognition (2026-10-06) → Paper 5: Epistemic Governance (2026-10-06) → Paper 6: Objective Transition (2026-10-06) → Paper 7: Survivability (2026-10-06) → **Paper 8: Resource-Constrained Intelligence (2026-10-06), this document**. Extensions recorded as binding on successors: the certificate lifecycle binds Paper 9's generated protocols (§11.2); the constraint taxonomy and per-module constraint objects bind Paper 10's domain modules (§11.2). Recorded inheritances: κ_k and the level lattice from Paper 7; the guard from Paper 6; the amendment path from Paper 5; bounded audit from Paper 4; the negative-space method from Paper 3; composition as meet from Paper 2; transition discipline from Paper 1; registry semantics from Paper 0.

12.5 Closing

With the feasibility engine the kernel gains its last input: not only what is permitted and what is supported, but what is affordable. The consequences are mundane and that is their value. An action outside the endowment returns no score rather than a bad one; a binding constraint reports its price rather than hiding it; an endowment measured at commissioning expires rather than coasting. Papers 9 through 11 inherit these predicates unchanged, and their work is easier for it.

— END OF PAPER 9 · SAVANT / CLAI-OS / AEGIS-GLOBAL · VERSION 1.0 · 2026-10-06 —

13 References

Internal artifacts

— SVT Architect Analysis — Constitutional Intelligence research and build program, 11 papers. Source architecture: Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL. Paper 8 specification: feasible

intervention space as five-fold intersection; argmax over the feasible set subject to safety; the rigorous-optimization standard for economic access.

— Paper 0: Constitutional Intelligence v1.0 (2026-09-30) — machine tuple, gates, consequence classes, registry semantics.

— Paper 1: Authority Transition Algebra v1.0 (2026-10-01) — transition objects and the algebra of lawful authority change.

— Paper 2: Constitutional Composition v1.0 (2026-10-02) — composition algebra, institutional coupling, meets and joins of constitutional objects.

— Paper 3: Negative-Space Intelligence v1.0 (2026-10-03) — failure-boundary calculus, absence classes, the registry of what did not happen.

— Paper 4: Bounded Metacognition v1.0 (2026-10-06) — recursive self-audit with operational stopping rules; the audit ladder.

— Paper 5: Epistemic Governance v1.0 (2026-10-06) — evidence–uncertainty–authority coupling, decay pricing, the amendment path.

— Paper 6: Objective Transition v1.0 (2026-10-06) — the guarded objective engine, interim mode, clinical objective-state transitions.

— Paper 7: Survivability v1.0 (2026-10-06) — the degradation lattice L0–L4, safety equivalence, deferred obligations, the paper tier.

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Appendices
Reference Material
The constraint taxonomy in full, and the certificate schemas.

A Constraint Taxonomy and Property Index

Table A-1 — Paper 8 objects, invariants, and obligations

ID	KIND	STATEMENT (ABBREVIATED)	STATUS
R-OBJ	Definition	Constraint record: predicate, endowment, consumption, decay, verification, custody (§2.3)	FH
Φ_R	Operator	Monotone feasibility operator with interaction clause (Eq 3-1)	FH
I-FE	Invariant	Only gfp(Φ_R) may be recommended, ranked, or emitted (§3.1)	FH
I-FC	Invariant	Feasibility-critical continuity: refusal, null output, proven escalation, record survive A_adm = ∅ (§3.3)	FH
FC-a/b/c/0	Certificate kinds	Assumed, cached, live, refusal — strength ordered, expiry enforced (§4.2)	FH
PoI	Certificate	Priced infeasibility: unconstrained argmax, binding constraint, utility gap, closing endowment (Eq 5-1)	FH
PO-9.1	Proof obligation	gfp(Φ_R) exists and is unique (Tarski; powerset lattice)	CONJ
PO-9.2	Proof obligation	Utility gap is a lower bound on true constraint cost (interaction monotonicity)	CONJ
PO-9.3	Proof obligation	Endowment enlargement weakly enlarges A_adm (monotonicity of relief)	CONJ
D-9.1	Design decision	Certificates are admission-time objects, never audit-time reconstructions (§10.2)	FH
H57–H63	Hypotheses	Table 10-1; benchmarks B-36–B-38	CONJ
κ_R calibration	Open parameter	Awaiting B-36–B-38; no defaults asserted	UNK

A.1 The six families, with their clocks

Table A-2 — Constraint families: clock, measurement, and custody

FAMILY	DOMINANT CLOCK	MEASUREMENT CADENCE	CUSTODY PRINCIPAL
Financial	Slow (budget cycles); fast at billing events	Per episode and per cycle	Institutional finance; payer contract
Temporal	Fast	Continuous — every decision carries its latency budget	The deployment itself
Workforce	Both — shift-level (fast), attrition (slow)	Per shift; per credential audit	Clinical governance; licensing bodies
Infrastructure	Both — Paper 7's two clocks verbatim	Per Paper 7's detection predicates	Facilities; Paper 7's level certification

Supply chain	Fast at consumption; slow at lead time	Per consumption event; per delivery custody event	Pharmacy / stores custody channel
Legal-licit	Slow — revocation and amendment events	Per renewal; per regulatory bulletin	Compliance office; external regulator

B Feasibility Certificate Schemas

The canonical certificate contents: what each kind attests, what grounds it, what ends it. Contents are defaults; deployment dossiers instantiate them per domain, alongside the survivability dossier of Paper 7.

Table B-1 — Certificate schemas: fields and standing rules

KIND	ATTESTS	REQUIRED FIELDS	STANDING RULES
FC-a	Design-time plausibility only	Constraint set; default source; assumption statement; correlated-auditor flag	One episode, non-renewable; lowest ceiling tier; every reliance logged
FC-b	Measured endowment within window	Constraint set; endowment floor; verification timestamp and agent; window expiry; consumption ledger ref	Expires at window end or floor-crossing; grounds standard authority at the current level
FC-c	Live, episode-scoped measurement	All FC-b fields, plus: measurement method; patient/plan binding; state-change triggers	Expires on any touching state change; grounds the highest authority the endowment class supports
FC-0	A constraint binds; refusal with the price	Binding constraint(s); deficit vs. floor; PoI fields (Eq 5-1); escalation path with its own feasibility proof	Never expires; aggregates into the deployment's PoI stream; routed per §5.4

Two invariants hold across every row. First, no certificate strengthens by passage of time: age only ever moves a certificate downward in kind, never upward — the resource reading of Paper 5's decay law. Second, no certificate survives its own falsification silently: expiry, floor-crossing, and state change are events in the constitutional record with the same standing as Paper 7's descent detections. The certificate system is the feasibility engine's memory, and like every memory in this architecture, it is forbidden to forget the intervals in which it was constrained.

Feasible optimality, as defined in §6, is a weaker claim than optimality and a stronger claim than most deployed systems can make: within a certified feasible set, the choice was optimal, and outside it, every exclusion has a certificate. The certificates are the deliverable. Audits, governance reviews, and the patients whose constraints were measured all read the same ledger.

Appendix RW — Related Work and Positioning

This appendix is part of the elevation pass of 2026-10. It positions the paper against adjacent research programmes; the main text's structure, numbering, and reference list are unchanged. In-text citations follow the paper's author-year style; new references are listed at the end of this appendix.

Constrained optimization. The mathematical spine is classical: a feasible set defined by hard constraints, an objective optimized inside it, and shadow prices reporting what each constraint costs. Koopmans's activity-analysis framework introduced the shadow price as the economic content of a binding constraint (Koopmans 1951); constrained Markov decision processes (Altman) and constrained policy optimization (Achiam et al.) bring the same structure to sequential decision-making under learned policies (Altman 1999; Achiam et al. 2017). Robust optimization (Ben-Tal and Nemirovski) contributes the posture toward the endowment itself: the feasible set is computed against an endowment that may be measured stale, and the price of that staleness is paid conservatively rather than averaged away (Ben-Tal and Nemirovski 2002). The paper's additions to this apparatus are two. First, infeasibility is a first-class constitutional event with a certificate, not a solver exit code: the price-of-infeasibility construct (§5) exists so that what the system cannot afford is recorded, priced, and auditable, rather than silently dropped from the output. Second, the feasible set has standing in the authority lattice: scarcity contracts authority through the same machinery (Papers 1, 3, 5) that contracts it for failed evidence or absent infrastructure.

The refusal of soft penalties. §1.4's rejection of penalty-based scarcity handling has direct support in the safe-RL record. Penalty and Lagrangian methods trade constraint violation against reward with a multiplier; under distribution shift or a mis-scaled objective, the trade goes the wrong way, which is why shielding-style hard enforcement (Alshiekh et al.) exists (Alshiekh et al. 2018). This paper takes the same side as the shielding literature, for the same reason, at the resource layer: an action outside the feasible set does not have a lower score; it is not available.

Fairness and allocation. §1.1 explains why the paper does not use the fairness vocabulary. That vocabulary does real work elsewhere: the documented record of resource-correlated harm in clinical systems (Obermeyer et al.'s analysis of a population-management algorithm's racial bias; Sjoding et al.'s pulse-oximetry findings; Martin et al. on polygenic-score portability) shows that allocation decisions embed population structure whether or not the system admits it (Obermeyer et al. 2019; Sjoding et al. 2020; Martin et al. 2019). The feasibility engine is not a fairness mechanism and does not claim to be one; it ensures that constraint violations are visible and priced. What it contributes to the fairness conversation is infrastructural: a system that cannot state its feasible set cannot be audited for how it allocates within one.

Deployment economics. The two-clock structure (consumption and erosion, §1.3) parallels the maintenance literature's distinction between operating cost and technical debt; Sculley et al.'s account of hidden technical debt in machine-learning systems documents how unmeasured slow erosion of a system's real operating conditions presents, eventually, as a sudden fast-clock failure (Sculley et al. 2015).

What this paper does not do. It does not reduce scarcity or claim to; the engine prices poverty, and a priced constraint is still a constraint. It does not make allocation decisions fair; feasibility

certificates record what was excluded, and equity review of those records is a governance act the engine enables but does not perform. It does not supply the endowment measurements; the slow clock's estimates come from deployment domains and are wrong until metered. It does not prove feasible optimality is achievable within the certificate format at scale; H57–H63 state the falsifiable content and are untested. And it does not cover adversarial manipulation of endowment reports; a deployment that lies about its wallet defeats the consumption clock, and the mitigation (signed telemetry, reconciliation against stock records) is specified as an obligation rather than demonstrated.

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